

Background: The Structure-Elucidation Challenge

Natural products, small molecules produced by plants, fungi, and marine organisms, have long served as a major source of therapeutic agents, with approximately **70% of approved drugs** derived from or inspired by natural product scaffolds [1]. Despite advances in compound discovery, structural elucidation remains a major challenge. Following isolation, the molecular structure of a compound must be determined primarily from nuclear magnetic resonance (NMR) spectra, a process that typically requires weeks of expert analysis and may extend to months or years for previously uncharacterized scaffolds. This work presents two complementary machine learning approaches to accelerate molecular structure prediction. **Part I** systematically evaluates the performance of representative generative architectures for molecular generation. **Part II** conditions molecular generation on experimental NMR spectra to predict candidate structures, including compounds absent from the reference database.

PART I · Comparing Molecule Generation Models

Molecule Representation

How a molecule is encoded determines what a model can learn, and how often it produces something chemically valid.

SMILES (text)

c1ccccc1

Compact and widely used. A single wrong character yields an invalid molecule, and random edits in text space rarely decode to valid chemistry.

Atom graphs

atoms + bonds

Preserves chemistry directly, but requires graph neural networks that keep the graph valid at every generation step.

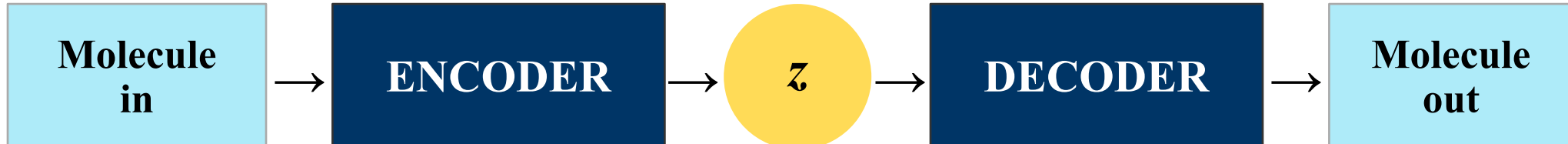
Motif graphs

tree of fragments [2]

Atoms are grouped into chemically meaningful fragments, rings and common scaffolds, decisions made per fragment rather than per character.

Architecture: Variational Autoencoders

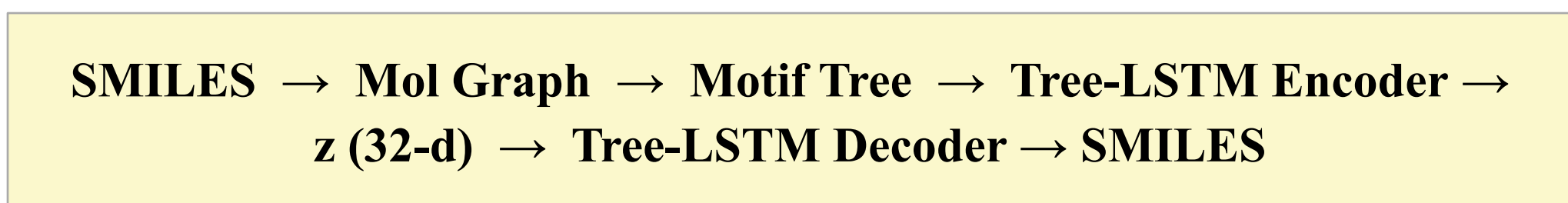
A VAE compresses a molecule into a latent vector $\mathbf{z} \in \mathbb{R}^d$ and rebuilds it with a decoder. Training pulls the latent distribution toward $N(0, I)$, turning that space into a smooth coordinate system over chemistry [3].



Generate: sample $\mathbf{z} \sim N(0, I)$, decode, obtain a new molecule.

Encode: map a known molecule to $\mathbf{z}$ for interpolation or property optimisation. The same weights serve both modes.

HierVAE (Jin et al., 2020) [4]



Three encoders - atom, motif, and tree level share weights, so structure is captured at three scales simultaneously.

Decoding is fragment-by-fragment: motifs are sampled greedily until the tree is complete, then assembled, so nearly all outputs are valid by construction.

The 32-d latent is deliberately narrow, which forces the model to encode only what is chemically significant.

Models Compared

Model	Approach	Year
CharRNN	LSTM over SMILES characters	2018
SMILES VAE	VAE on tokenised SMILES	2018
AAE	AE + GAN-style latent regulariser	2018
LatentGAN	GAN over a pretrained AE	2019
JT-VAE	VAE over a tree of fragments	2018
HierVAE ★	Hierarchical motif VAE	2020
GMT-SELFIES	Transformer on SELFIES	2023

Dataset & Metrics

All models were trained and scored on **MOSES** [5], a standard benchmark of drug-like molecules, through one identical evaluation pipeline.

- **Validity:** parseable by RDKit.
- **Uniqueness at 10k:** distinct molecules per 10,000 samples.
- **Novelty:** absent from the training set.
- **Internal Diversity:** mean pairwise Tanimoto distance.
- **Reconstruction:** exact-match accuracy after encode–decode.

Results: Generation Benchmarks

Model	Valid	Uniqueness at 10k	Novelty	Internal Diversity
JT-VAE	1.000	0.999	0.997	0.860
CharRNN	0.981	0.949	0.783	0.860
SMILES VAE	0.816	0.973	0.916	0.861
AAE	0.912	0.967	0.899	0.860
SELFIES-VAE	1.000	0.935	1.000	0.873
HierVAE @32	1.000	0.811	0.993	0.849
HierVAE @256	1.000	0.811	0.993	0.851

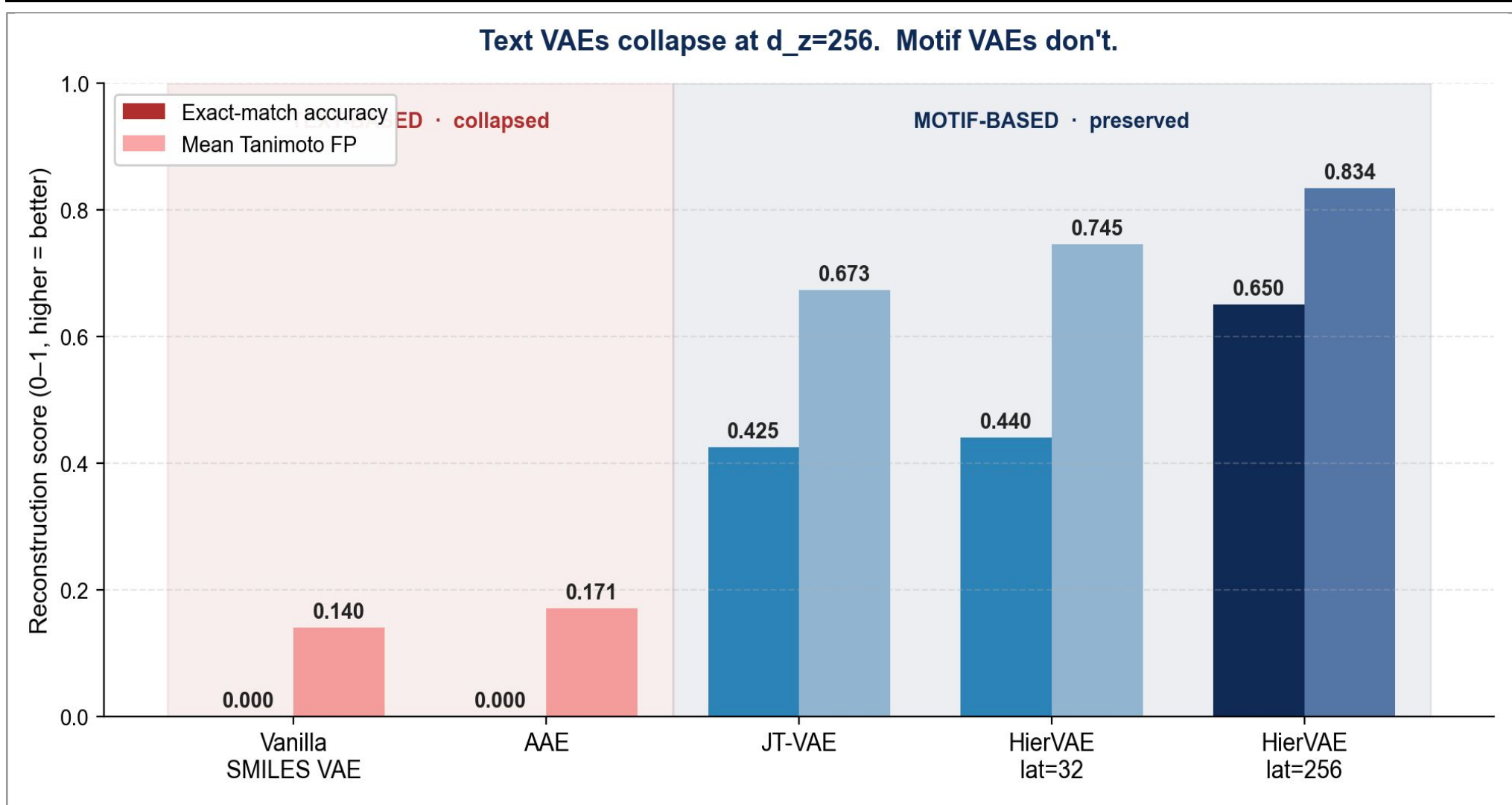
Sampling quality for every model trained in this study. Motif-based decoders (JT-VAE, HierVAE) reach perfect validity; text-based decoders trade validity for sample uniqueness.

Experiment 1: JT-VAE vs HierVAE

Metric	JT-VAE (56-d)	HierVAE (32-d)
Uniqueness at 10k	0.999	0.802
Novelty	0.997	0.993
Internal diversity	0.860	0.849
Reconstruction	0.425	0.440
Mean Tanimoto	0.673	0.745

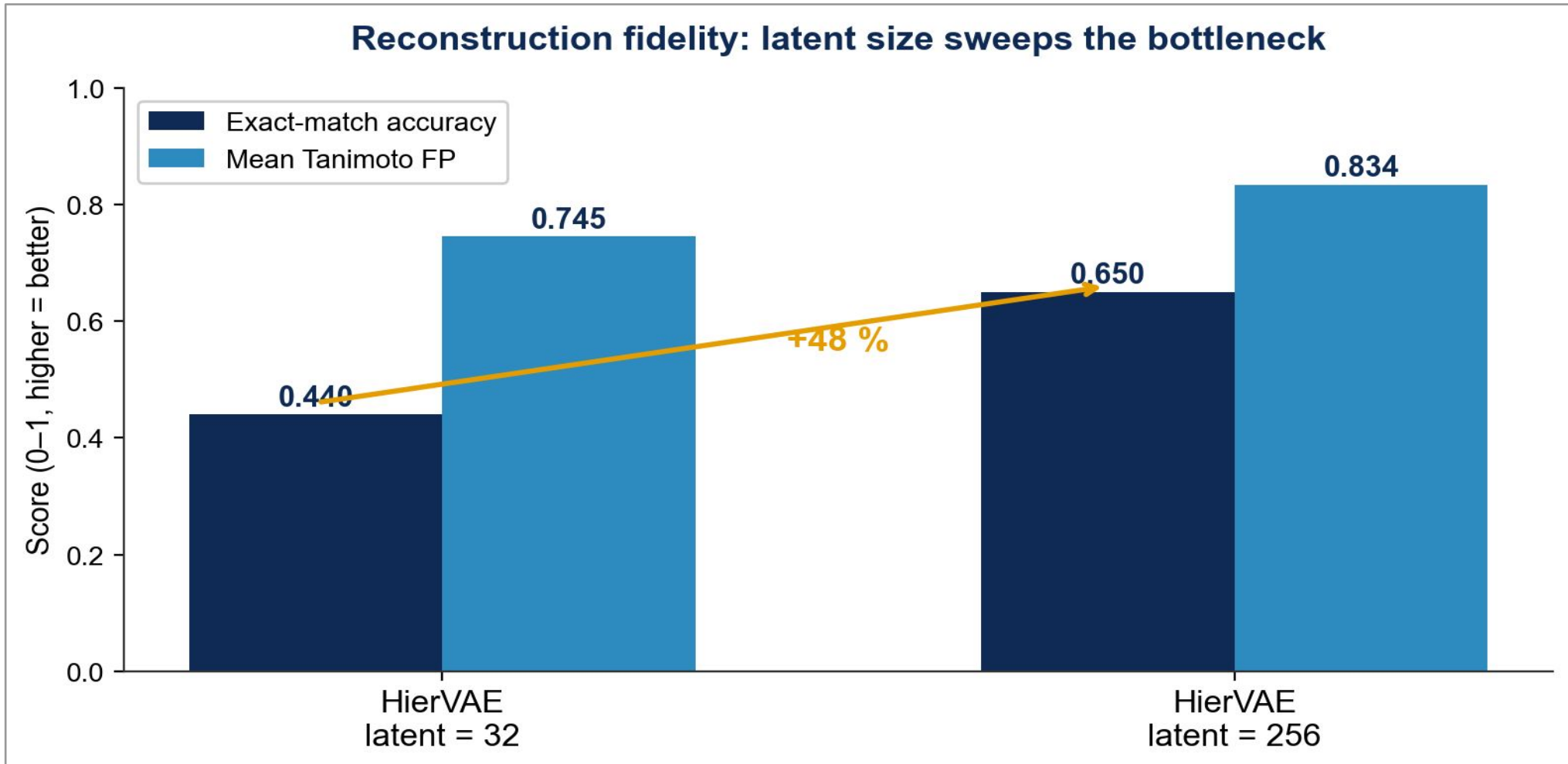
Both architectures at their published latent sizes, evaluated identically. HierVAE wins on reconstruction fidelity, JT-VAE on sample uniqueness.

Reconstruction: Text vs. Motif



Molecular reconstruction measured by accuracy and mean Tanimoto similarity. At 256-d, text-based VAEs reach zero exact matches, while motif-based models retain reconstruction capacity.

Effect of Latent Width



Widening HierVAE's latent from 32 to 256 dimensions lifts exact-match reconstruction from 0.440 to 0.650, a 48% relative gain, and Tanimoto from 0.745 to 0.834.

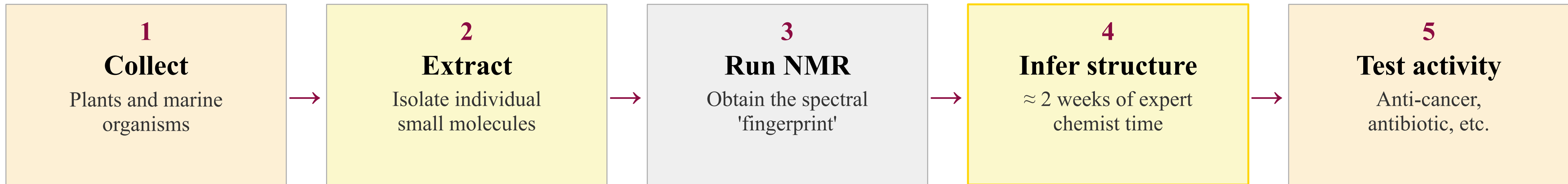
Findings - Part I

Representation dominates architecture. Motif-based models achieved 100% validity without post-processing; text-based VAEs fell to 81.6%.

Decoder structure prevents collapse. Wider latent spaces degraded text-based VAEs but preserved motif-based performance.

Latent width is a tunable trade-off. 256-d space improved reconstruction, while 32-d favored novel sampling.

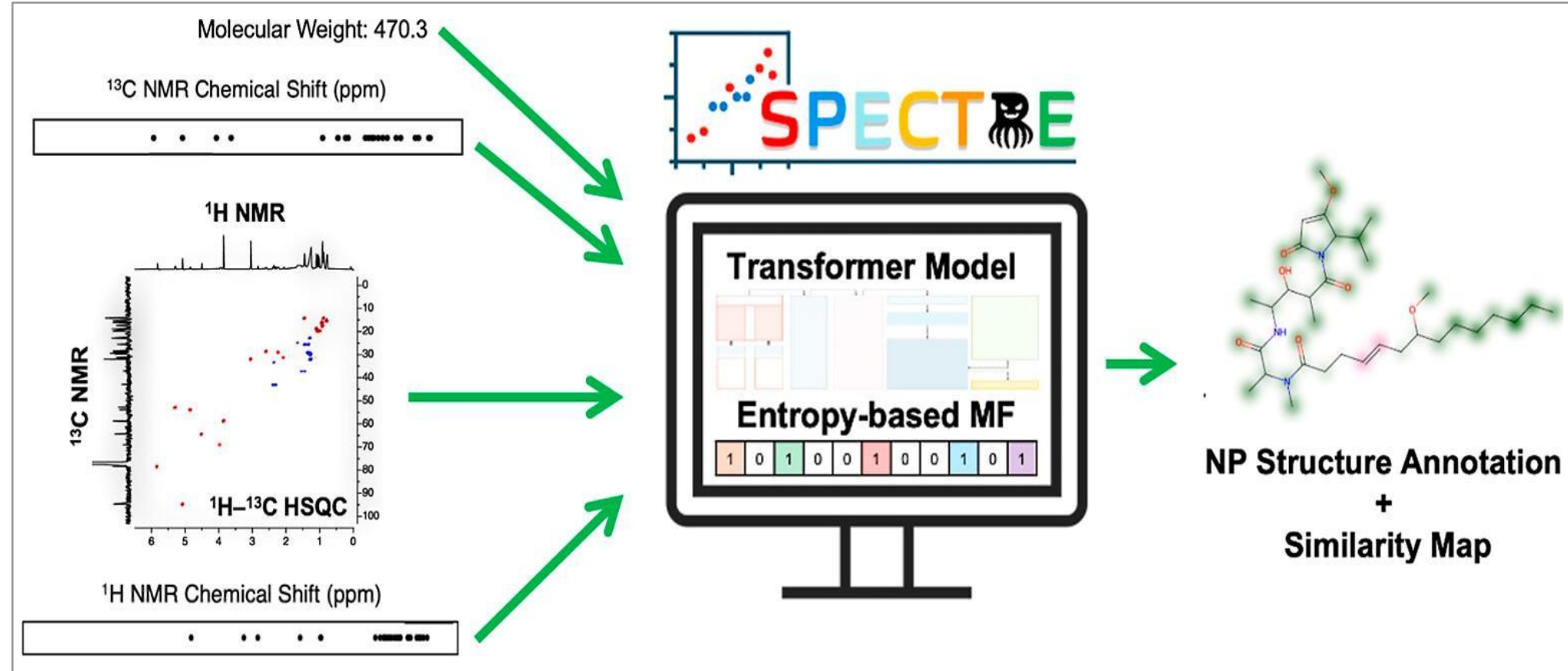
THE NATURAL PRODUCT DISCOVERY PIPELINE



PART II · HIGHT-Conditioned Graph Diffusion Pipeline

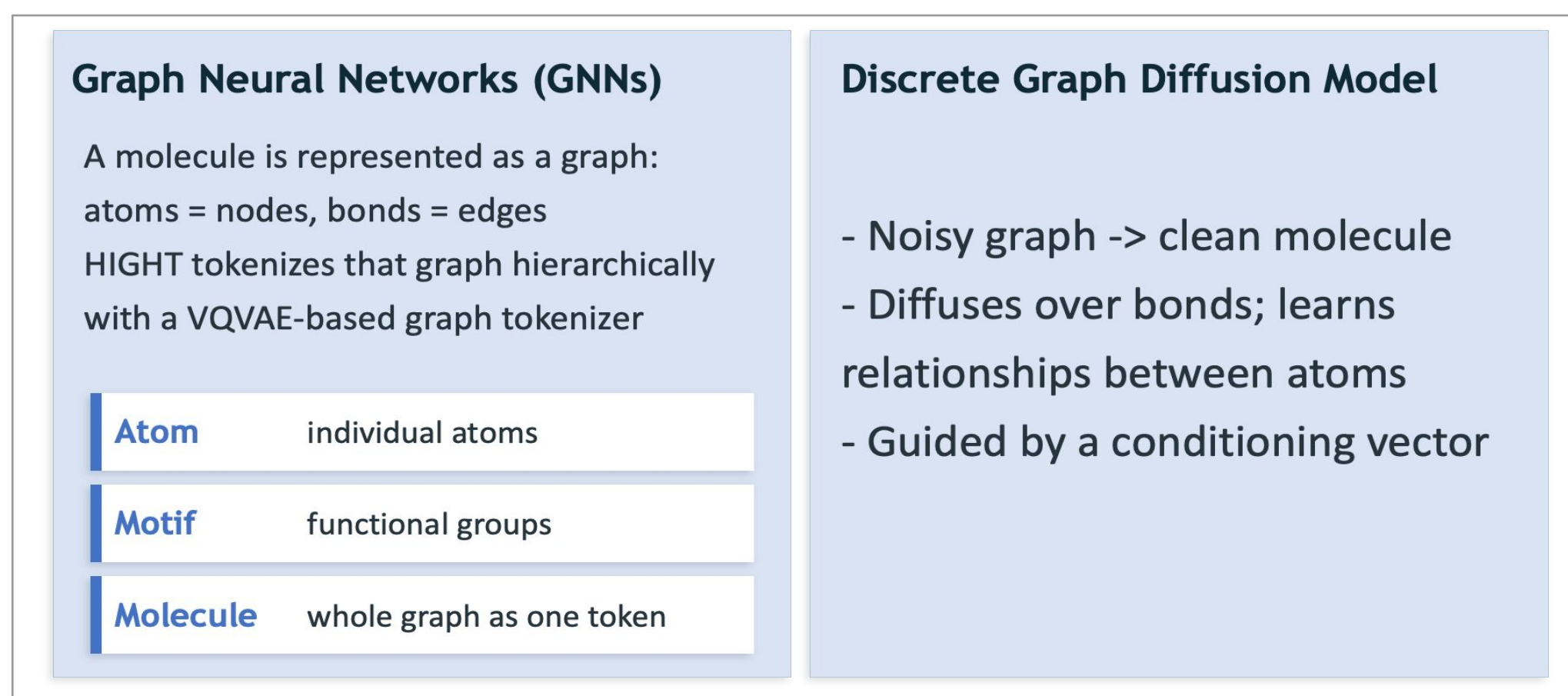
The Challenge: Novel Structures

Determining the structure of a newly isolated natural product from its NMR spectra is a time intensive step in discovery. MARINA/SPECTRE [6] addresses this by retrieval, embedding the spectrum and searching a database of known structures for the closest match. However, exact retrieval requires the target structure to be present in that library.



MARINA/SPECTRE encodes NMR spectra and molecular weight into an entropy-based Morgan fingerprint, used to annotate structure by similarity search.

Approach: HIGHT + DiffMS



HIGHT tokenizes a molecular graph hierarchically (atom, motif, molecule); DiffMS denoises a noisy graph into a clean molecule under a conditioning vector.

HIGHT (Hierarchical Graph Tokenization) [7] encodes retrieved candidate structures at three levels of granularity. **DiffMS** (Diffusion Generation of Molecules Conditioned on Mass Spectra) [8] is a discrete graph diffusion model that builds a molecular graph from a conditioning vector.

→ Integrating the two incorporates structural information from retrieved candidates without restricting generation to the reference library.

Research Question

Can a HIGHT-conditioned diffusion model propose valid natural-product structures that do not exist in the retrieval database?

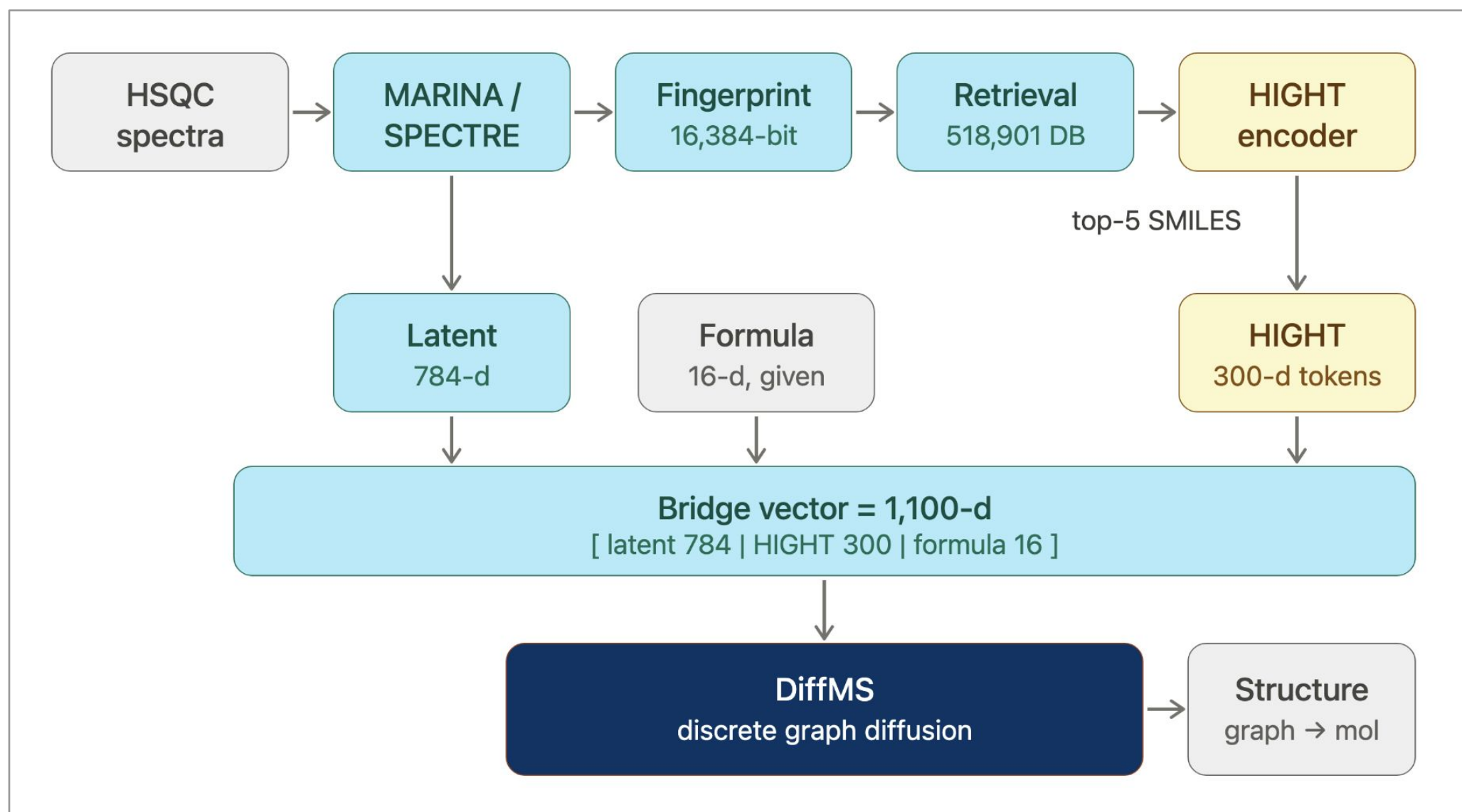
Evaluation: Test Set

150 molecules were drawn from the DiffMS test split, structures the model never saw in training. Each target was excluded from the reference library before retrieving the *five* conditioning candidates. Any exact recovery was therefore generated rather than retrieved, while novelty was assessed against all five candidates.

Metrics

- **Validity:** the output graph converts to an RDKit-parseable molecule.
- **Tanimoto similarity:** fingerprint similarity between the generated structure and the withheld ground truth.
- **Novelty:** the generated SMILES differs from all five retrieved candidates.

Pipeline



Three stages. MARINA/SPECTRE turns HSQC spectra into a 16,384-bit fingerprint and a 784-d latent; retrieval returns the top-5 SMILES, which HIGHT encodes into 300-d tokens; the concatenated 1,100-d bridge vector conditions DiffMS.

Three Stages

Stage 1 - MARINA / SPECTRE

HSQC spectra in. Outputs a 16,384-bit Morgan fingerprint for retrieval and a 784-d latent embedding for the bridge vector.

Stage 2 - HIGHT

Takes the top-5 candidate SMILES from cosine retrieval and encodes them at atom, motif, and molecule level into a 300-d representation.

Stage 3 - DiffMS

A GraphTransformer denoiser consumes the 1,100-d bridge vector (784 latent + 300 HIGHT + 16 formula) and emits a molecular graph, converted to an RDKit molecule.

Results

Tanimoto Similarity	Resembles Target	Novel (unlike Top-5)
≥ 0.3	29 (19.3%)	15 (10.0%)
≥ 0.4	15 (10.0%)	4 (2.7%)
≥ 0.6	7 (4.7%)	0 (0%)

Similarity of generated structures to the withheld ground truth, and how many were novel.

4-epoch model, 150 test molecules. **98 (65.3%)** produced a valid single molecule, **33 (22.0%)** a valid but disconnected graph, and **19 (12.7%)** nothing chemically valid. Of the valid outputs, 29 (19.3%) reached Tanimoto ≥ 0.3 against the withheld truth, and 15 of those were also novel.

Findings - Part II

The pipeline runs end-to-end at 65.3% single-molecule validity after only 4 epochs.

It generates rather than retrieves. 15 of 150 targets (10.0%) yielded a structure that both resembles the withheld truth (Tanimoto ≥ 0.3) and differs from every retrieved candidate.

Next Steps

Train substantially longer and run ablations that remove the HIGHT tokens, the SPECTRE latent, and the formula in turn, to measure what each contributes to the bridge vector.

Key References

- [1] Newman & Cragg. Natural products as sources of new drugs, 1981–2019. J Nat Prod 83:770 (2020).
- [2] Jin, Barzilay & Jaakkola. Junction tree VAE for molecular graph generation. ICML (2018).
- [3] Kingma & Welling. Auto-encoding variational Bayes. ICLR (2014).
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- [5] Polykovskiy et al. Molecular Sets (MOSES): benchmarking molecular generation. Front Pharmacol 11:565644 (2020).
- [6] Xu et al. SPECTRE: a multimodal spectral transformer for small molecule annotation. J Chem Inf Model (2025).
- [7] Chen et al. HIGHT: hierarchical graph tokenization for molecule–language alignment. arXiv:2406.14021 (2024).
- [8] Bohde et al. DiffMS: diffusion generation of molecules conditioned on mass spectra. ICML (2025).